

Lecture 10 Optimizations

Recap on Lagrange's Multiplier:

$$\max/\min \quad \text{objective function} \quad \underline{f}(x, y, z) \quad \text{s.t.} \quad \underline{g}(x, y, z) = c \quad \text{constraints}$$

1. Solve the equations
$$\begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) \\ g(x, y, z) = c \end{cases}$$

Here the unknowns are x, y, z and λ

2. The solutions of (x, y, z) are possible candidates of the boundary critical points maximum/minimum points

3. Evaluate f at each point. The largest/smallest value of f is the maximum/minimum.

Ex. Find the distance from $(0, 0, 0)$ to the plane $2x + 3y + 4z = 29$.

Sol. A point (x, y, z) on the plane $2x + 3y + 4z = 29$

$$\text{minimize } x^2 + y^2 + z^2 \quad \text{s.t. } 2x + 3y + 4z = 29$$

$$f(x, y, z) = x^2 + y^2 + z^2 \quad g(x, y, z) = 2x + 3y + 4z$$

$$\begin{cases} 2x = 2\lambda \\ 2y = 3\lambda \\ 2z = 4\lambda \\ 2x + 3y + 4z = 29 \end{cases}$$

$$\begin{cases} x = \lambda \\ y = \frac{3}{2}\lambda \\ z = 2\lambda \end{cases} \Rightarrow 2\lambda + \frac{9}{2}\lambda + 8\lambda = 29 \Rightarrow \lambda = 2$$

$$\Rightarrow (x, y, z) = (2, 3, 4)$$

$$\Rightarrow \text{distance} = \sqrt{x^2 + y^2 + z^2} = \sqrt{29}$$

Ex. Many airlines require that the sum of length, width and height of a checked baggage cannot exceed 62 inches. Find the dimensions of the rectangular baggage that has the greatest possible volume.

Sol. l, w, h

$$\max V(l, w, h) = lwh \quad \text{s.t.} \quad l + w + h = 62 \quad g(l, w, h) = l + w + h$$

$$\begin{cases} wh = \lambda \\ lh = \lambda \\ lw = \lambda \\ l + w + h = 62 \end{cases} \quad (l, w, h) = \left(\frac{62}{3}, \frac{62}{3}, \frac{62}{3} \right), (0, 0, 62) \\ (0, 62, 0), (62, 0, 0)$$

$$\Rightarrow \max V = \left(\frac{62}{3} \right)^3.$$

Ex. Three cities A, B, and C are located at $(5, 2)$, $(-4, 4)$ and $(-1, -3)$ respectively. There is a railtrack whose equation is $y = x^3 + 1$, and a station is going to be built on the track so that the sum of squares of the distances from each city to the station is minimized. Find the coordinates of the station.

Sol. Suppose the station is at (x, y)

$$\min f(x, y) = (x-5)^2 + (y-2)^2 + (x+4)^2 + (y-4)^2 + (x+1)^2 + (y+3)^2$$

$$\text{s.t.} \quad g(x, y) = y - x^3 = 1$$

$$\begin{cases} \nabla f = \lambda \nabla g \\ g(x, y) = 1 \end{cases} \Rightarrow \begin{cases} 2(x-5) + 2(x+4) + 2(x+1) = -3\lambda x^2 \\ 2(y-2) + 2(y-4) + 2(y+3) = \lambda \\ y - x^3 = 1 \end{cases}$$

$$\Rightarrow \begin{cases} 6x = -3\lambda x^2 \\ 6y - 6 = \lambda \\ y - x^3 = 1 \end{cases}$$

$$\text{Case 1: } x = 0 \Rightarrow (x, y) = (0, 1)$$

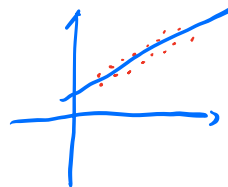
$$\text{Case 2: } x \neq 0 \Rightarrow \begin{cases} x = -\frac{2}{\lambda} \\ y = 1 + \frac{2}{\lambda} \end{cases} \Rightarrow 1 + \frac{2}{\lambda} + \frac{8}{\lambda^3} = 1 \\ \Rightarrow \frac{2}{\lambda} = -\frac{8}{\lambda^3} \Rightarrow \lambda^2 = -48.$$

$$\Rightarrow \text{Minimum distance} = f(0, 1)$$

Ex. (Least Square Approximation). Given a set of data points $(x_1, y_1), \dots, (x_N, y_N)$ on the xy -plane.

Find the straight line $y = mx + c$ such that the sum of squares of distances between each (x_i, y_i) and $(x_i, mx_i + c)$ is minimized. Linear Regression

Sol. $\min f(m, c) = \sum_{i=1}^N (y_i - mx_i - c)^2$



$$\frac{\partial f}{\partial m} = -2 \sum_{i=1}^N (y_i - mx_i - c) x_i$$

$$= -2 \left(\sum_{i=1}^N x_i y_i - m \sum_{i=1}^N x_i^2 - c \sum_{i=1}^N x_i \right) = 0$$

$$\frac{\partial f}{\partial c} = -2 \sum_{i=1}^N (y_i - mx_i - c) = -2 \left(\sum_{i=1}^N y_i - m \sum_{i=1}^N x_i - cN \right) = 0$$

$$\Rightarrow \begin{cases} m \sum_{i=1}^N x_i^2 + c \sum_{i=1}^N x_i = \sum_{i=1}^N x_i y_i \\ m \sum_{i=1}^N x_i + cN = \sum_{i=1}^N y_i \end{cases}$$

$$\Rightarrow \begin{cases} m = \frac{(\sum x_i)(\sum y_i) - N(\sum x_i y_i)}{(\sum x_i)^2 - N(\sum x_i^2)} \\ c = \frac{(\sum x_i)(\sum x_i y_i) - (\sum x_i^2)(\sum y_i)}{(\sum x_i)^2 - N(\sum x_i^2)} \end{cases}$$

Ex. $f(x, y) = x^2 - 4x + y^2 + 9$. Find the absolute maximum/minimum of f restricted to the domain $4x^2 + 9y^2 \leq 36$.

Sol. 1° Interior

$$\text{set } \nabla f = \vec{0} \quad \begin{cases} 2x - 4 = 0 \\ 2y = 0 \end{cases} \Rightarrow (2, 0).$$

2° Boundary.

$$\min/\max f(x,y) = x^2 - 4x + y^2 + 9 \quad \text{s.t.} \quad g(x,y) = 4x^2 + 9y^2 = 36.$$

$$\begin{cases} 2x - 4 = 8\lambda x \\ 2y = 18\lambda y \\ 4x^2 + 9y^2 = 36. \end{cases}$$

$$\text{Case 1. } y \neq 0 \Rightarrow \lambda = \frac{1}{9} \Rightarrow 2x - 4 = \frac{8}{9}x \Rightarrow x = \frac{18}{5}$$

$$9y^2 = 36 - 4\left(\frac{18}{5}\right)^2 < 0.$$

$$\text{Case 2. } y=0 \Rightarrow 4x^2 = 36 \Rightarrow x = \pm 3$$

$$\Rightarrow (3,0), (-3,0).$$

$$3^\circ \quad f(2,0) = 5 \quad \text{absolute minimum}$$

$$f(3,0) = 6$$

$$f(-3,0) = 30 \quad \text{absolute maximum.}$$

Procedure. Finding Absolute Maximum/Minimum Values on Closed, Bounded Sets,

f is defined on a closed bounded set R on $\mathbb{R}^2$

1. Determine the values of f at all critical points in R
2. Find the maximum and minimum values of f on the boundary of R
3. The greatest/least function value found in step 1 and 2 is the absolute